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\\Research\Investigators\XChen\Keto MRS 20240709\localizer

TA: 0:19 PM: REF Voxel size: 0.5×0.5×10.0 mmPAT: Off Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	On
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	5
Dist. factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	2
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	3
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	280 mm
FoV phase	100.0 %
Slice thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	7
Filter	Prescan Normalize, Elliptical filter
Coil elements	HC1-7

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0 ms
MTC	Off
Magn. preparation	None
Flip angle	40 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Contrast - Dynamic

Averages	1
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	280 mm
FoV phase	100.0 %
Slice thickness	10.0 mm
Base resolution	256
Phase resolution	50 %
Phase partial Fourier	Off
Interpolation	On

Resolution - iPAT

PAT mode	None

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	On

Geometry - Common

Slice group	1
Slices	5
Dist. factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	2
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	3
Slices	1
Dist. factor	20 %
Position	Isocenter
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	280 mm
FoV phase	100.0 %
Slice thickness	10.0 mm
TR	20.0 ms
Multi-slice mode	Sequential
Series	Interleaved
Concatenations	7

Geometry - AutoAlign

Slice group	1
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	2
Position	Isocenter

Geometry - AutoAlign

Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	3
Position	Isocenter
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Rotation	0.00 deg
Initial Orientation	Sagittal

Geometry - Saturation

Saturation mode	Standard
Fat suppr.	None
Water suppr.	None
Special sat.	None

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Tune up
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	20.0 ms
Concatenations	7
Segments	1

Physio - Cardiac

Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	280 mm
FoV phase	100.0 %
Phase resolution	50 %

Physio - PACE

Resp. control	Off
Concatenations	7

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Liver registration	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Soft Tissue

Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off
Measurements	1

Inline - Composing

Distortion Corr.	Off

Sequence - Part 1

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Sequential
Bandwidth	180 Hz/Px

Sequence - Part 2

Segments	1
Acoustic noise reduction	None
RF pulse type	Fast
Gradient mode	Normal
Excitation	Slice-sel.
RF spoiling	On

Sequence - Assistant

Mode	Off
Allowed delay	0 s

\\Research\Investigators\XChen\Keto MRS 20240709\AAHScout
TA: 0:17 PM: REF Voxel size: 1.6×1.6×1.6 mmPAT: 2 Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	On
Load images to graphic segments	On
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slab group	1
Slabs	1
Dist. factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Phase oversampling	0 %
Slice oversampling	0.0 %
Slices per slab	128
FoV read	260 mm
FoV phase	100.0 %
Slice thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
Filter	None
Coil elements	HC1-7

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip angle	8 deg

Contrast - Dynamic

Averages	1
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1

Resolution - Common

FoV read	260 mm
FoV phase	100.0 %
Slice thickness	1.6 mm
Base resolution	160
Phase resolution	100 %
Slice resolution	69 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Trajectory	Cartesian

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Accel. factor 3D	1

Resolution - iPAT

Reference scan mode	Integrated

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slab group	1
Slabs	1
Dist. factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice oversampling	0.0 %
Slices per slab	128
FoV read	260 mm
FoV phase	100.0 %
Slice thickness	1.6 mm
TR	3.15 ms
Multi-slice mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab group	1
Position	Isocenter
Orientation	Sagittal
Phase enc. dir.	A >> P
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Rotation	0.00 deg
Initial Orientation	Transversal

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Tune up
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off

System - Adjustments

Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	Low
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Flip angle	8 deg
Measurements	1
Time to center	7.5 s

Inline - Inline

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off

Sequence - Part 1

Introduction	On
Dimension	3D
Asymmetric echo	Weak
Contrasts	1
Multi-slice mode	Sequential
Bandwidth	540 Hz/Px

Sequence - Part 2

RF pulse type	Fast
Gradient mode	Normal
Excitation	Non-sel.
RF spoiling	On

Sequence - Assistant

Mode	Off

\\Research\Investigators\XChen\Keto MRS 20240709\T1_MEMPRAGE_64ch
TA: 6:02 PM: REF Voxel size: 1.0×1.0×1.0 mmPAT: 2 Rel. SNR: 1.00 : tfl_me

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	On
Start measurements	Single measurement

Routine

Slab group	1
Slabs	1
Dist. factor	50 %
Position	R1.8 A13.5 F34.8 mm
Orientation	S > C1.4 > T-0.9
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Phase oversampling	0 %
Slice oversampling	0.0 %
Slices per slab	176
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	1.00 mm
TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
Filter	Raw filter, Prescan Normalize
Coil elements	HC1-7;NC1,2

Contrast - Common

TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. preparation	Non-sel. IR
TI	1100 ms
Flip angle	7.0 deg
Fat suppr.	None
Water suppr.	None

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %

Resolution - Common

Slice thickness	1.00 mm
Base resolution	256
Phase resolution	100 %
Slice resolution	100 %
Phase partial Fourier	Off
Slice partial Fourier	Off
Interpolation	Off

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	32
Accel. factor 3D	1
Reference scan mode	Integrated

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	On
Elliptical filter	Off

Geometry - Common

Slab group	1
Slabs	1
Dist. factor	50 %
Position	R1.8 A13.5 F34.8 mm
Orientation	S > C1.4 > T-0.9
Phase enc. dir.	A >> P
Slice oversampling	0.0 %
Slices per slab	176
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	1.00 mm
TR	2530.0 ms
Multi-slice mode	Single shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab group	1
Position	R1.8 A13.5 F34.8 mm
Orientation	S > C1.4 > T-0.9
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L1.0 A13.6 F32.1
L	1.0 mm
A	13.6 mm
F	32.1 mm
Initial Rotation	5.58 deg
Initial Orientation	Sagittal

Geometry - Navigator

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R1.8 A13.5 F34.8 mm
Orientation	S > C1.4 > T-0.9
Rotation	-7.80 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	Low
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2530.0 ms
Concatenations	1

Physio - Cardiac

Magn. preparation	Non-sel. IR
TI	1100 ms
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off

Sequence - Part 1

Introduction	Off
Dimension	3D
Elliptical scanning	Off
Reordering	Linear
Asymmetric echo	Off
Contrasts	4
Flow comp. 1	No
Multi-slice mode	Single shot
Echo spacing	9.8 ms
Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px
Bandwidth 4	650 Hz/Px

Sequence - Part 2

RF pulse type	Fast
Gradient mode	Fast
Excitation	Non-sel.
RF spoiling	On
Incr. Gradient spoiling	Off
Turbo factor	176

Sequence - Special

Readout polarity	Positive
Readout trajectory	Bipolar
Gradient spoiling	Siemens
Gradient moment factor	1
Averaging	RMS

Sequence - Assistant

Mode	Off

\\Research\Investigators\XChen\Keto MRS 20240709\t2_tse_dark-fluid_tra
TA: 4:14 PM: REF Voxel size: 0.7×0.7×4.0 mmPAT: Off Rel. SNR: 1.00 : tir

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	On
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	33
Dist. factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	220 mm
FoV phase	96.9 %
Slice thickness	4.0 mm
TR	9000.0 ms
TE	83.0 ms
Averages	1
Concatenations	2
Filter	Prescan Normalize, Elliptical filter
Coil elements	HC1-7;NC1,2

Contrast - Common

TR	9000.0 ms
TE	83.0 ms
TD	0.0 ms
MTC	Off
Magn. preparation	Slice-sel. IR
TI	2500 ms
Flip angle	150 deg
Fat suppr.	Fat sat.
Fat sat. mode	Strong
Water suppr.	None
Restore magn.	Off
Freeze suppressed tissue	On

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	220 mm
FoV phase	96.9 %
Slice thickness	4.0 mm
Base resolution	320
Phase resolution	70 %

Resolution - Common

Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off

Resolution - iPAT

PAT mode	None

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	On

Geometry - Common

Slice group	1
Slices	33
Dist. factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	R >> L
FoV read	220 mm
FoV phase	96.9 %
Slice thickness	4.0 mm
TR	9000.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	2

Geometry - AutoAlign

Slice group	1
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L1.6 A6.8 F2.6
L	1.6 mm
A	6.8 mm
F	2.6 mm
Initial Rotation	89.56 deg
Initial Orientation	T > C
T > C	-9.8
> S	0.0

Geometry - Saturation

Fat suppr.	Fat sat.
Fat sat. mode	Strong
Water suppr.	None
Restore magn.	Off
Special sat.	None

Geometry - Navigator

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Rotation	90.90 deg
R >> L	214 mm
A >> P	220 mm
F >> H	164 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	9000.0 ms
Concatenations	2

Physio - Cardiac

Magn. preparation	Slice-sel. IR
TI	2500 ms
Fat suppr.	Fat sat.
Dark blood	Off
FoV read	220 mm
FoV phase	96.9 %
Phase resolution	70 %
Trajectory	Cartesian

Physio - PACE

Resp. control	Off
Concatenations	2

Inline - Common

Subtract	Off

Inline - Common

Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off

Sequence - Part 1

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	On
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	8.28 ms
Bandwidth	289 Hz/Px

Sequence - Part 2

Define	Turbo factor
Echo trains per slice	13
Phase correction	Automatic
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Performance
Hyperecho	Off
WARP	Off
Red. EC sensitivity	Off
Turbo factor	17

Sequence - Assistant

Mode	Off
Allowed delay	60 s

\\Research\Investigators\XChen\Keto MRS 20240709\pd+t2_tse_tra

TA: 2:11 PM: FIX Voxel size: 0.3×0.3×4.0 mmPAT: 2 Rel. SNR: 1.00 : tse

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	33
Dist. factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	220 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
TR	3600.0 ms
TE 1	9.4 ms
TE 2	94 ms
Averages	1
Concatenations	1
Filter	Prescan Normalize, Elliptical filter
Coil elements	HC1-7;NC1,2

Contrast - Common

TR	3600.0 ms
TE 1	9.4 ms
TE 2	94 ms
MTC	Off
Magn. preparation	None
Flip angle	160 deg
Fat suppr.	None
Water suppr.	None
Restore magn.	Off

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	220 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
Base resolution	320
Phase resolution	100 %
Phase partial Fourier	Off
Trajectory	Cartesian

Resolution - Common

Interpolation	On

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	30
Reference scan mode	Integrated

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	On

Geometry - Common

Slice group	1
Slices	33
Dist. factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	R >> L
FoV read	220 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
TR	3600.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L1.6 A6.8 F2.6
L	1.6 mm
A	6.8 mm
F	2.6 mm
Initial Rotation	89.56 deg
Initial Orientation	T > C
T > C	-9.8
> S	0.0

Geometry - Saturation

Fat suppr.	None
Water suppr.	None
Restore magn.	Off
Special sat.	None

Geometry - Navigator

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Tune up
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3600.0 ms
Concatenations	1

Physio - Cardiac

Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	220 mm
FoV phase	100.0 %
Phase resolution	100 %
Trajectory	Cartesian

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Distortion Corr.	Off

Sequence - Part 1

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	On
Contrasts	2
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	9.36 ms
Bandwidth	252 Hz/Px

Sequence - Part 2

Define	Turbo factor
Echo trains per slice	35
Phase correction	Automatic
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Normal
Hyperecho	Off
WARP	Off
Red. EC sensitivity	Off
Turbo factor	5

Sequence - Assistant

Mode	Min flip angle
Min flip angle	130 deg
Allowed delay	60 s

\\Research\Investigators\XChen\Keto MRS 20240709\resolve_4scan_trace_tra_p2_192

TA: 1:51 PM: FIX Voxel size: 1.1×1.1×4.0 mmPAT: 2 Rel. SNR: 1.00 : resolve

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	30
Dist. factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	220 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
TR	4040 ms
TE 1	55 ms
TE 2	93 ms
Concatenations	1
Filter	Raw filter, Prescan Normalize
Coil elements	HC1-7;NC1,2

Contrast - Common

TR	4040 ms
TE 1	55 ms
TE 2	93 ms
MTC	Off
Magn. preparation	None
Flip angle	180 deg
Fat suppr.	Fat sat.
Fat sat. mode	Strong

Contrast - Dynamic

Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1

Resolution - Common

FoV read	220 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
Base resolution	192
Phase resolution	100 %
Phase partial Fourier	Off
Readout partial Fourier	5/8
Readout segments	7
Interpolation	Off

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	96
Reference scan mode	EPI/separate

Resolution - Filter Image

Distortion Corr.	Off
Prescan Normalize	On
Unfiltered images	Off

Resolution - Filter Rawdata

Raw filter	On

Geometry - Common

Slice group	1
Slices	30
Dist. factor	25 %
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	A >> P
FoV read	220 mm
FoV phase	100.0 %
Slice thickness	4.0 mm
TR	4040 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L1.6 A6.8 F2.6
L	1.6 mm
A	6.8 mm
F	2.6 mm
Initial Rotation	-0.44 deg
Initial Orientation	T > C
T > C	-9.8
> S	0.0

Geometry - Saturation

Fat suppr.	Fat sat.
Fat sat. mode	Strong
Special sat.	None

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	Head > Brain

System - Miscellaneous

Coil Select Mode	Off - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R1.6 A13.7 F4.6 mm
Orientation	T > C3.6 > S0.9
Rotation	0.90 deg
A >> P	220 mm
R >> L	220 mm
F >> H	149 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	4040 ms
Concatenations	1

Diff - Neuro

Diffusion mode	4-Scan Trace
Diff. directions	4
Diffusion Scheme	Monopolar
Diff. weightings	2
b-value 1	0 s/mm²
b-value 2	1000 s/mm²
b-value 1	1
b-value 2	1
Diff. weighted images	Off
Trace weighted images	On
ADC maps	On
FA maps	Off
Mosaic	Off
Tensor	Off
Noise level	100

Diff - Body

Diffusion mode	4-Scan Trace
Diff. directions	4
Diffusion Scheme	Monopolar
Diff. weightings	2
b-value 1	0 s/mm²
b-value 2	1000 s/mm²
b-value 1	1
b-value 2	1
Diff. weighted images	Off

Diff - Body

Trace weighted images	On
ADC maps	On
Exponential ADC Maps	Off
FA maps	Off
Invert Gray Scale	Off
Calculated Image	Off
b-Value >=	0 s/mm²
Noise level	100

Diff - Composing

Distortion Corr.	Off

Sequence - Part 1

Introduction	On
Dimension	2D
Contrasts	2
Optimization	Min. TE
Multi-slice mode	Interleaved
Echo spacing	0.32 ms
Bandwidth	766 Hz/Px

Sequence - Part 2

EPI factor	96
RF pulse type	Normal
Gradient mode	Fast
Reacquisition mode	On

Sequence - Assistant

Mode	Off

\\Research\Investigators\XChen\Keto MRS 20240709\slicepositioning
TA: 0:20 PM: FIX Voxel size: 2.0×2.0×2.0 mmPAT: 2 Rel. SNR: 1.00 : epfid

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	On
Start measurements	Single measurement

Routine

Slice group	1
Slices	70
Dist. factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	216 mm
FoV phase	100.0 %
Slice thickness	2.00 mm
TR	800 ms
TE	25.00 ms
Averages	1
Multi-band accel. factor	5
Filter	Prescan Normalize
Coil elements	HC1-7

Contrast - Common

TR	800 ms
TE	25.00 ms
MTC	Off
Magn. preparation	None
Flip angle	52 deg
Fat suppr.	Fat sat.

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Delay in TR	0 ms

Resolution - Common

FoV read	216 mm
FoV phase	100.0 %
Slice thickness	2.00 mm
Base resolution	108
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	32

Resolution - iPAT

Reference scan mode	Single-shot

Resolution - Filter Image

Distortion Corr.	Off
Prescan Normalize	On

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off
Hamming	Off

Geometry - Common

Slice group	1
Slices	70
Dist. factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
FoV read	216 mm
FoV phase	100.0 %
Slice thickness	2.00 mm
TR	800 ms
Multi-slice mode	Interleaved
Series	Interleaved
Multi-band accel. factor	5

Geometry - AutoAlign

Slice group	1
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R4.5 A7.2 H1.2
R	4.5 mm
A	7.2 mm
H	1.2 mm
Initial Rotation	0.00 deg
Initial Orientation	T > C
T > C	-20.0
> S	0.0

Geometry - Saturation

Fat suppr.	Fat sat.
Special sat.	None

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Sum of Squares
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Standard

System - Adjustments

B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Rotation	1.49 deg
A >> P	216 mm
R >> L	216 mm
F >> H	140 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	800 ms
Multi-band accel. factor	5

BOLD

GLM Statistics	Off
Dynamic t-maps	Off
Ignore meas. at start	0
Ignore after transition	0
Model transition states	On
Temp. highpass filter	On
Threshold	4.00
Paradigm size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion correction	Off
Spatial filter	Off

BOLD

Measurements	1
Delay in TR	0 ms

Sequence - Part 1

Introduction	Off
Dimension	2D
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	0.56 ms
Bandwidth	2314 Hz/Px

Sequence - Part 2

EPI factor	108
Gradient mode	Performance*
Excitation	Standard
RF spoiling	Off

Sequence - Special

Excite pulse duration	4380 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.0 deg
Fat sat. offset	0.0 Hz
Sinc exc. pulse BWTP	5.2
Physio recording	DICOM
Triggering scheme	Standard

\\Research\Investigators\XChen\Keto MRS 20240709\restingstate_mb5_AP
TA: 8:19 PM: FIX Voxel size: 2.0×2.0×2.0 mmPAT: 2 Rel. SNR: 1.00 : epfid

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	On
Wait for user to start	On
Start measurements	Single measurement

Routine

Slice group	1
Slices	70
Dist. factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	216 mm
FoV phase	100.0 %
Slice thickness	2.00 mm
TR	800 ms
TE	25.00 ms
Averages	1
Multi-band accel. factor	5
Filter	Prescan Normalize
Coil elements	HC1-7

Contrast - Common

TR	800 ms
TE	25.00 ms
MTC	Off
Magn. preparation	None
Flip angle	52 deg
Fat suppr.	Fat sat.

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	600
Delay in TR	0 ms

Resolution - Common

FoV read	216 mm
FoV phase	100.0 %
Slice thickness	2.00 mm
Base resolution	108
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	32

Resolution - iPAT

Reference scan mode	Single-shot

Resolution - Filter Image

Distortion Corr.	Off
Prescan Normalize	On

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off
Hamming	Off

Geometry - Common

Slice group	1
Slices	70
Dist. factor	0 %
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
FoV read	216 mm
FoV phase	100.0 %
Slice thickness	2.00 mm
TR	800 ms
Multi-slice mode	Interleaved
Series	Interleaved
Multi-band accel. factor	5

Geometry - AutoAlign

Slice group	1
Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R4.5 A7.2 H1.2
R	4.5 mm
A	7.2 mm
H	1.2 mm
Initial Rotation	0.00 deg
Initial Orientation	T > C
T > C	-20.0
> S	0.0

Geometry - Saturation

Fat suppr.	Fat sat.
Special sat.	None

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Sum of Squares
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Standard

System - Adjustments

B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R7.8 A14.8 F1.0 mm
Orientation	T > C-6.6 > S0.7
Rotation	1.49 deg
A >> P	216 mm
R >> L	216 mm
F >> H	140 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	800 ms
Multi-band accel. factor	5

BOLD

GLM Statistics	Off
Dynamic t-maps	Off
Ignore meas. at start	0
Ignore after transition	0
Model transition states	On
Temp. highpass filter	On
Threshold	4.00
Paradigm size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion correction	Off
Spatial filter	Off

BOLD

Measurements	600
Delay in TR	0 ms

Sequence - Part 1

Introduction	Off
Dimension	2D
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	0.56 ms
Bandwidth	2314 Hz/Px

Sequence - Part 2

EPI factor	108
Gradient mode	Performance*
Excitation	Standard
RF spoiling	Off

Sequence - Special

Excite pulse duration	4380 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.0 deg
Fat sat. offset	0.0 Hz
Sinc exc. pulse BWTP	5.2
Physio recording	DICOM
Triggering scheme	Standard

\\Research\Investigators\XChen\Keto MRS 20240709\dMRI_dir64_MGH_AP
TA: 3:48 PM: FIX Voxel size: 1.5×1.5×1.5 mmPAT: Off Rel. SNR: 1.00 : epse

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	On
Start measurements	Single measurement

Routine

Slice group	1
Slices	92
Dist. factor	0 %
Position	R3.2 A10.1 H1.6 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Phase oversampling	0 %
FoV read	210 mm
FoV phase	100.0 %
Slice thickness	1.50 mm
TR	3230 ms
TE	89.20 ms
Multi-band accel. factor	4
Filter	None
Coil elements	HC1-7

Contrast - Common

TR	3230 ms
TE	89.20 ms
MTC	Off
Magn. preparation	None
Flip angle	78 deg
Refocus flip angle	160 deg
Fat suppr.	None
Grad. rev. fat suppr.	Enabled

Contrast - Dynamic

Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Delay in TR	0 ms
Multiple series	Off

Resolution - Common

FoV read	210 mm
FoV phase	100.0 %
Slice thickness	1.50 mm
Base resolution	140
Phase resolution	100 %
Phase partial Fourier	6/8
Interpolation	Off

Resolution - iPAT

PAT mode	None

Resolution - Filter Image

Distortion Corr.	Off
Prescan Normalize	Off
Dynamic Field Corr.	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	92
Dist. factor	0 %
Position	R3.2 A10.1 H1.6 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
FoV read	210 mm
FoV phase	100.0 %
Slice thickness	1.50 mm
TR	3230 ms
Multi-slice mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice group	1
Position	R3.2 A10.1 H1.6 mm
Orientation	T > C-6.6 > S0.7
Phase enc. dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A1.9 H2.6
R	0.0 mm
A	1.9 mm
H	2.6 mm
Initial Rotation	0.00 deg
Initial Orientation	T > C
T > C	-20.0
> S	0.0

Geometry - Saturation

Fat suppr.	None
Grad. rev. fat suppr.	Enabled
Special sat.	None

Geometry - Navigator

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Sum of Squares
Matrix Optimization	Off
AutoAlign	Head > Brain
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Standard

System - Adjustments

B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R3.2 A10.1 H1.6 mm
Orientation	T > C-6.6 > S0.7
Rotation	1.49 deg
A >> P	210 mm
R >> L	210 mm
F >> H	138 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3230 ms
Multi-band accel. factor	4

Physio - PACE

Resp. control	Off
Multi-band accel. factor	4

Diff - Neuro

Diffusion mode	MDDW
Diff. directions	64
Diffusion Scheme	Monopolar
Diff. weightings	2
b-value 1	0 s/mm²
b-value 2	2000 s/mm²
b-value 1	1
b-value 2	1
Diff. weighted images	On
Trace weighted images	On
ADC maps	On
FA maps	On
Mosaic	On
Tensor	On
Noise level	40

Diff - Body

Diffusion mode	MDDW
Diff. directions	64
Diffusion Scheme	Monopolar
Diff. weightings	2
b-value 1	0 s/mm²
b-value 2	2000 s/mm²
b-value 1	1
b-value 2	1

Diff - Body

Diff. weighted images	On
Trace weighted images	On
ADC maps	On
Exponential ADC Maps	Off
FA maps	On
Invert Gray Scale	Off
Calculated Image	Off
b-Value >=	0 s/mm²
Noise level	40

Diff - Composing

Distortion Corr.	Off

Sequence - Part 1

Introduction	On
Dimension	2D
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	0.69 ms
Bandwidth	1700 Hz/Px

Sequence - Part 2

EPI factor	140
Gradient mode	Performance
Excitation	Standard
RF spoiling	Off

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	DICOM

\\Research\Investigators\XChen\Keto MRS 20240709\fastestmap
TA: 6.1 s PM: REF Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI F >> H	25 mm
VoI A >> P	22 mm
TR	2000 ms
TE	49.20 ms
Averages	1
Filter	None
Coil elements	HC3-6

Contrast

TR	2000 ms
TE	49.20 ms
Tau	5.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI R >> L	33 mm
VoI A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T

System - Miscellaneous

Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2000 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Linear 3-proj
VoI fit factor	100 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm
Inversion pulse	Off
Multi-echo acquisition	On

Sequence - Special

Number of echoes	16

\\Research\Investigators\XChen\Keto MRS 20240709\fastestmap
TA: 0:12 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	2000 ms
TE	49.20 ms
Averages	1
Filter	None
Coil elements	HC3-6

Contrast

TR	2000 ms
TE	49.20 ms
Tau	5.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T

System - Miscellaneous

Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2000 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
Vol fit factor	100 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm
Inversion pulse	Off
Multi-echo acquisition	On

Sequence - Special

Number of echoes	16

\\Research\Investigators\XChen\Keto MRS 20240709\fastestmap
TA: 0:12 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	2000 ms
TE	49.20 ms
Averages	1
Filter	None
Coil elements	HC3-6

Contrast

TR	2000 ms
TE	49.20 ms
Tau	10.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T

System - Miscellaneous

Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2000 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
Vol fit factor	100 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm
Inversion pulse	Off
Multi-echo acquisition	On

Sequence - Special

Number of echoes	16

\\Research\Investigators\XChen\Keto MRS 20240709\fastestmap

TA: 0:12 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI F >> H	25 mm
VoI A >> P	22 mm
TR	2000 ms
TE	49.20 ms
Averages	1
Filter	None
Coil elements	HC3-6

Contrast

TR	2000 ms
TE	49.20 ms
Tau	10.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI R >> L	33 mm
VoI A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2000 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
VoI fit factor	100 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm
Inversion pulse	Off
Multi-echo acquisition	On

Sequence - Special

Number of echoes	16

\\Research\Investigators\XChen\Keto MRS 20240709\fastestmap
TA: 0:12 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : fastmp

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI F >> H	25 mm
VoI A >> P	22 mm
TR	2000 ms
TE	49.20 ms
Averages	1
Filter	None
Coil elements	HC3-6

Contrast

TR	2000 ms
TE	49.20 ms
Tau	10.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
Measurements	1

Resolution - Common

Vector size	256

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI R >> L	33 mm
VoI A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T

System - Miscellaneous

Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	2000 ms

Sequence - Common

Delta frequency	0.0 ppm
Phase cycling	None
Bandwidth	100000 Hz
Acquisition duration	2 ms

Sequence - Special

Type of fit	Linear 6-proj
VoI fit factor	100 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.0 mm
Inversion pulse	Off
Multi-echo acquisition	On

Sequence - Special

Number of echoes	16

\\Research\Investigators\XChen\Keto MRS 20240709\svs_slaser_dkd_WSFA
TA: 0:33 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : sead

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Averages	1
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Total TE	28 ms
Averages	1
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
VAPOR WS	Enabled
Water suppr. BW	70 Hz
VAPOR Flip angle	60 deg

Resolution - Common

Prescan Normalize	Off
Vector size	2048

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg

Geometry - AutoAlign

Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Save single averages	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Preparation scans	2
Delta frequency	0.0 ppm
Water Ref. Scan	Off
Phase cycling	Auto
Bandwidth	3000 Hz
Acquisition duration	682 ms
Remove oversampling	On

Sequence - Special

Excitation duration	2000 us

Sequence - Special

Refocusing duration	4000 us
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
OVS module 1	On
OVS module 2	On
OVS module 3	On
OVS module 4	On
OVS slab X	120 mm
OVS slab Y up	120 mm
OVS slab Y down	120 mm
OVS slab Z	120 mm
OVS gap	7 mm
OVS delay 7	83 ms
OVS delay 8	21 ms
Calibration Type	VAPOR WS FA
Debug Type	None
Increment step size	10 deg
Measurements	9
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\XChen\Keto MRS 20240709\svs_slaser_dkd
TA: 3:18 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : sead

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI F >> H	25 mm
VoI A >> P	22 mm
TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Averages	64
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Total TE	28 ms
Averages	64
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
VAPOR WS	Enabled
Water suppr. BW	70 Hz
VAPOR Flip angle	110 deg

Resolution - Common

Prescan Normalize	Off
Vector size	2048

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI R >> L	33 mm
VoI A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg

Geometry - AutoAlign

Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Save single averages	On
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Preparation scans	2
Delta frequency	0.0 ppm
Water Ref. Scan	Off
Phase cycling	Auto
Bandwidth	3000 Hz
Acquisition duration	682 ms
Remove oversampling	On

Sequence - Special

Excitation duration	2000 us

Sequence - Special

Refocusing duration	4000 us
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
OVS module 1	On
OVS module 2	On
OVS module 3	On
OVS module 4	On
OVS slab X	120 mm
OVS slab Y up	120 mm
OVS slab Y down	120 mm
OVS slab Z	120 mm
OVS gap	7 mm
OVS delay 7	80 ms
OVS delay 8	21 ms
Calibration Type	None
Debug Type	None
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\XChen\Keto MRS 20240709\svs_slaser_dkd_EC
TA: 0:30 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : sead

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Averages	8
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Total TE	28 ms
Averages	8
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
VAPOR WS	Only RF off

Resolution - Common

Prescan Normalize	Off
Vector size	2048

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Save single averages	On
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Preparation scans	2
Delta frequency	0.0 ppm
Phase cycling	Auto
Bandwidth	3000 Hz
Acquisition duration	682 ms
Remove oversampling	On

Sequence - Special

Excitation duration	2000 us
Refocusing duration	4000 us
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
OVS module 1	On

Sequence - Special

OVS module 2	On
OVS module 3	On
OVS module 4	On
OVS slab X	120 mm
OVS slab Y up	120 mm
OVS slab Y down	120 mm
OVS slab Z	120 mm
OVS gap	7 mm
OVS delay 7	83 ms
OVS delay 8	21 ms
Calibration Type	None
Debug Type	None
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\XChen\Keto MRS 20240709\svs_slaser_dkd_wq
TA: 0:30 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : sead

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Averages	8
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE 1	8 ms
TE 2	11 ms
TE 3	9 ms
Total TE	28 ms
Averages	8
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
VAPOR WS	None

Resolution - Common

Prescan Normalize	Off
Vector size	2048

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Save single averages	On
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Preparation scans	2
Delta frequency	0.0 ppm
Phase cycling	Auto
Bandwidth	3000 Hz
Acquisition duration	682 ms
Remove oversampling	On

Sequence - Special

Excitation duration	2000 us
Refocusing duration	4000 us
GOIA/FOCI	Off
Inversion pulse	Off
FA AutoCalib	Off
Calibration Type	None

Sequence - Special

Debug Type	None
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\XChen\Keto MRS 20240709\eja_svs_mpress_ws_rACC
TA: 0:41 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : mpres

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	3000 ms
TE	68.00 ms
Averages	1
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE	68.00 ms
Averages	1
Excite flip angle	90 deg
Refocus flip angle	180 deg
VAPOR	Enabled
VAPOR suppr.	Water suppr.
Water s. BW	110 Hz
Water s. delta pos.	0.00 ppm
Measurements	9

Resolution - Common

Vector size	1024

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX

System - Miscellaneous

Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Brain
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	On
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Introduction	On
Preparation scans	2
Delta frequency	-1.7 ppm
Phase cycling	Auto
Bandwidth	2000 Hz
Acquisition duration	512 ms
Remove oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.0 mT/m
Refocus grad. factor	1.00
OVS slab thickness	80.0 mm
OVS slab pos. offset	5.0 mm

Sequence - Special

Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	100 deg
VAPOR delay 8	16 ms
VAPOR delay 7	60 ms
OVS pulse duration	5120 us
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Weighted averages	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
MEGA water suppr.	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	WS FA
WS FA inc.	10 deg
Measurements	9
Number of label freqs.	1
Editing pulse freq. [1]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\XChen\Keto MRS 20240709\eja_svs_mpress_4p56Am25D1200bw54
TA: 13:14 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : mpres

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	3000 ms
TE	68.00 ms
Averages	128
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE	68.00 ms
Averages	128
Excite flip angle	90 deg
Refocus flip angle	180 deg
VAPOR	Enabled
VAPOR suppr.	Water suppr.
Water s. BW	110 Hz
Water s. delta pos.	0.00 ppm
Measurements	1

Resolution - Common

Vector size	1024

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX

System - Miscellaneous

Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	On
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Introduction	On
Preparation scans	2
Delta frequency	-1.7 ppm
Phase cycling	Auto
Bandwidth	2000 Hz
Acquisition duration	512 ms
Remove oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.0 mT/m
Refocus grad. factor	1.00
OVS slab thickness	80.0 mm
OVS slab pos. offset	5.0 mm

Sequence - Special

Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	160 deg
VAPOR delay 8	16 ms
VAPOR delay 7	65 ms
OVS pulse duration	5120 us
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	On
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
MEGA water suppr.	On
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	7.50 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\XChen\Keto MRS 20240709\eja_svs_mpress_rACC_EC
TA: 0:38 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : mpres

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI F >> H	25 mm
VoI A >> P	22 mm
TR	3000 ms
TE	68.00 ms
Averages	4
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE	68.00 ms
Averages	4
Excite flip angle	90 deg
Refocus flip angle	180 deg
VAPOR	Only RF off
VAPOR suppr.	Water suppr.
Water s. BW	110 Hz
Water s. delta pos.	0.00 ppm
Measurements	1

Resolution - Common

Vector size	1024

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI R >> L	33 mm
VoI A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX

System - Miscellaneous

Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	On
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Introduction	On
Preparation scans	2
Delta frequency	-1.7 ppm
Phase cycling	Auto
Bandwidth	2000 Hz
Acquisition duration	512 ms
Remove oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.0 mT/m
Refocus grad. factor	1.00
OVS slab thickness	80.0 mm
OVS slab pos. offset	5.0 mm

Sequence - Special

Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	130 deg
VAPOR delay 8	16 ms
VAPOR delay 7	60 ms
OVS pulse duration	5120 us
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	On
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	7.50 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\XChen\Keto MRS 20240709\eja_svs_mpress_rACC_wq

TA: 0:38 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : mpres

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI F >> H	25 mm
VoI A >> P	22 mm
TR	3000 ms
TE	68.00 ms
Averages	4
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE	68.00 ms
Averages	4
Excite flip angle	90 deg
Refocus flip angle	180 deg
VAPOR	None
VAPOR suppr.	Water suppr.
Water s. BW	110 Hz
Water s. delta pos.	0.00 ppm
Measurements	1

Resolution - Common

Vector size	1024

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
VoI F >> H	25 mm
VoI R >> L	33 mm
VoI A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

System - Miscellaneous

Positioning mode	FIX

System - Miscellaneous

Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
AutoAlign	---
Coil Select Mode	Off - All

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adj. water suppr.	Off
Adjust with body coil	On
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Sequence - Common

Introduction	On
Preparation scans	2
Delta frequency	-1.7 ppm
Phase cycling	Auto
Bandwidth	2000 Hz
Acquisition duration	512 ms
Remove oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.0 mT/m
Refocus grad. factor	1.00
Spoiler duration	1200 us
Acq. window shift	200 us

Sequence - Special

Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
Enable OVS	Off
Resolve averages	On
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	7.50 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators\XChen\Keto MRS 20240709\svs_se_lactate
TA: 13:00 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : svs_se

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	3000 ms
TE	270 ms
Averages	256
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE	270 ms
Averages	256
Flip angle	90 deg
Water suppr.	Water sat.
Water suppr. BW	50 Hz
Spectral suppr.	None
Measurements	1

Resolution - Common

Prescan Normalize	Off
Vector size	1024

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

Geometry - Navigator

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Save single averages	On
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adj. water suppr.	On
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Physio - PACE

Resp. control	Off

Sequence - Common

Preparation scans	4
Delta frequency	-3.3 ppm
Ref. scan mode	Off
Phase cycling	Auto
Bandwidth	2000 Hz
Acquisition duration	512 ms
Remove oversampling	On

\\Research\Investigators\XChen\Keto MRS 20240709\svs_se_lactate_WS
TA: 1:00 PM: FIX Vol: 33 ×25 ×22 mmRel. SNR: 1.00 : svs_se

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol F >> H	25 mm
Vol A >> P	22 mm
TR	3000 ms
TE	270 ms
Averages	16
Filter	None
Coil elements	HC3-6

Contrast

TR	3000 ms
TE	270 ms
Averages	16
Flip angle	90 deg
Water suppr.	Only RF off
Water suppr. BW	50 Hz
Spectral suppr.	None
Measurements	1

Resolution - Common

Prescan Normalize	Off
Vector size	1024

Geometry - Common

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	0 deg
Vol F >> H	25 mm
Vol R >> L	33 mm
Vol A >> P	22 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R4.4 A69.9 H7.7
R	4.4 mm
A	69.9 mm
H	7.7 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

Geometry - Navigator

System - Miscellaneous

Positioning mode	FIX
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Save single averages	On
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
B1 Shim mode	TrueForm
Adj. water suppr.	On
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	R4.4 A69.9 H7.7 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	25 mm
R >> L	33 mm
A >> P	22 mm
Reset	Off

System - pTx Volumes

B1 Shim mode	TrueForm

System - Tx/Rx

Frequency 1H	123.249501 MHz
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	3000 ms

Physio - PACE

Resp. control	Off

Sequence - Common

Preparation scans	4
Delta frequency	0.0 ppm
Ref. scan mode	Off
Phase cycling	Auto
Bandwidth	2000 Hz
Acquisition duration	512 ms
Remove oversampling	On